The DFT Circular Convolution Property

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Key Concepts

1) Multiplication of length-N DFT coefficients corresponds to convolution with an N-periodic extension of one of the signals and is called circular convolution.

2) In general circular convolution is not equal to ordinary or linear convolution.

    a) Circular convolution is equal to a sum of shifted (by integer multiples of N) replicates of the linear convolution.

    b) The linear convolution can only be obtained if the shifted replicates in the circular convolution do not overlap.

c) The condition for no overlap between replicates is that N be greater than or equal to the sum of the durations of the signals minus one.

Video Screenshots

DFT Convolution Property

Convolution $\overset{\text{DFT}}{\underset{?}{\longleftrightarrow}}$ Multiplication

Consider $\tilde{Y}[k] = X[k]H[k]$ where $x[n] \overset{\text{DFT}_N}{\longleftrightarrow} X[k]$, $h[n] \overset{\text{DFT}_N}{\longleftrightarrow} H[k]$

$$\tilde{y}[n] = \frac{1}{N}\sum_{k=0}^{N-1} H[k]X[k]e^{j\frac{2\pi}{N}kn}$$

$$= \frac{1}{N}\sum_{k=0}^{N-1}\left(\sum_{m=0}^{N-1} h[m]e^{-j\frac{2\pi}{N}km}\right)X[k]e^{j\frac{2\pi}{N}kn}$$

$$= \sum_{m=0}^{N-1} h[m]\left[\left(\frac{1}{N}\sum_{k=0}^{N-1} X[k]e^{j\frac{2\pi}{N}kn}\right)e^{-j\frac{2\pi}{N}km}\right]$$

where $\left(\frac{1}{N}\sum_{k=0}^{N-1} X[k]e^{j\frac{2\pi}{N}kn}\right) = x[((n))_N]$, and the bracketed term $= x[((n-m))_N]$

$$= \sum_{m=0}^{N-1} h[m]\,x[((n-m))_N]$$

$$\tilde{Y}[k] = H[k]X[k] \quad \overset{DFT_N}{\longleftrightarrow} \quad \tilde{y}[n] = \sum_{m=0}^{N-1} h[m]\,x[((n-m))_N]$$

$$= h[n] \circledast x[n]$$

$\circledast$ "circular convolution": convolve $h[n]$ with an N-periodic extension of $x[n]$

$$h[n] \circledast x[n] \overset{DFT_N}{\longleftrightarrow} H[k]X[k]$$

In general —

$$h[n] \circledast x[n] \neq h[n] * x[n]$$

Since $x[((n))_N] = \sum_{\ell=-\infty}^{\infty} x[n-\ell N]$

$$\tilde{y}[n] = h[n] \circledast x[n] = h[n] * x[((n))_N]$$

$$= h[n] * \sum_{\ell=-\infty}^{\infty} x[n-\ell N]$$

$$= \sum_{\ell=-\infty}^{\infty} h[n] * x[n-\ell N] \quad \text{let } y[n] = h[n] * x[n]$$

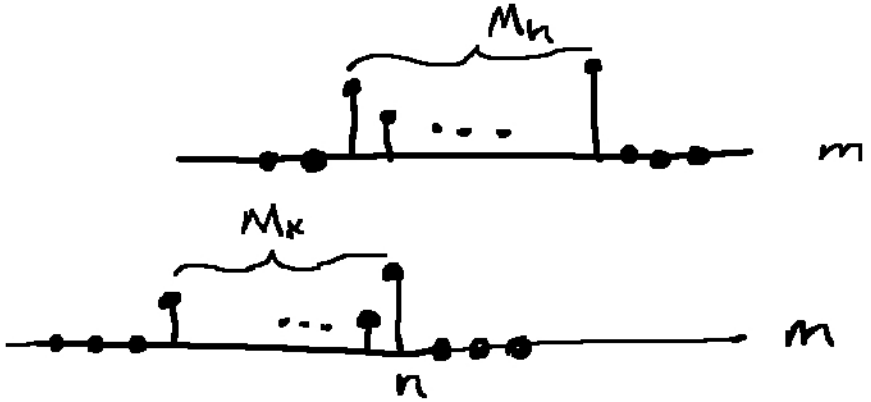
$$= \sum_{\ell=-\infty}^{\infty} y[n-\ell N]$$

sum of shifted replicates

Shifted replicates overlap if duration $y[n] > N$

Can obtain $y[n]$ from $\tilde{y}[n]$ iff duration $y[n] \leq N$

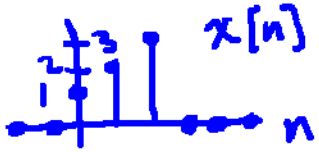
If $x[n]$ is duration M_x and $h[n]$ is duration M_h,
then $y[n] = h[n] * x[n]$ is duration $M_y = M_x + M_h - 1$



Choose $N \geq M_x + M_h - 1$ to obtain $y[n] = h[n] * x[n]$

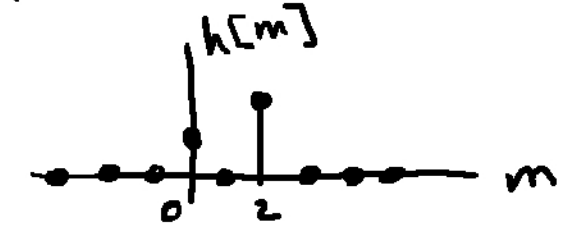
from $\tilde{y}[n] = h[n] \textcircled{N} x[n] \xleftrightarrow{DFT;N} H[k]X[k]$

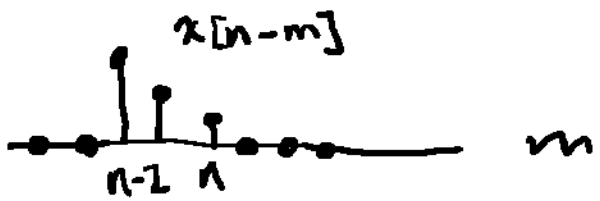
Example:





$$y[n] = h[n] * x[n] = \sum_m h[m]x[n-m]$$





$y[n] = 0, \quad n < 0$

$y[0] = 1$

$y[1] = 2$

$y[2] = 5$

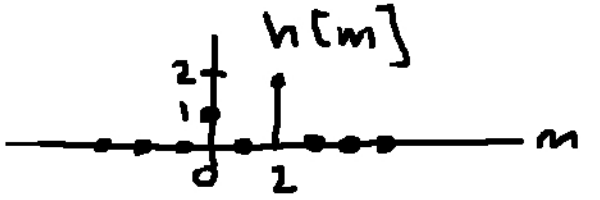
$y[3] = 4$

$y[4] = 6$

$y[n] = 0 \quad n \geq 5$

$\tilde{y}[n] = h[n] \textcircled{N} x[n], \quad N = 3$

$= h[n] * x[((n))_3]$





$\tilde{y}[n] \neq 0 \quad n < 0$

$\tilde{y}[0] = 5$

$\tilde{y}[1] = 8$

$\tilde{y}[2] = 5$

$\tilde{y}[3] = 5$

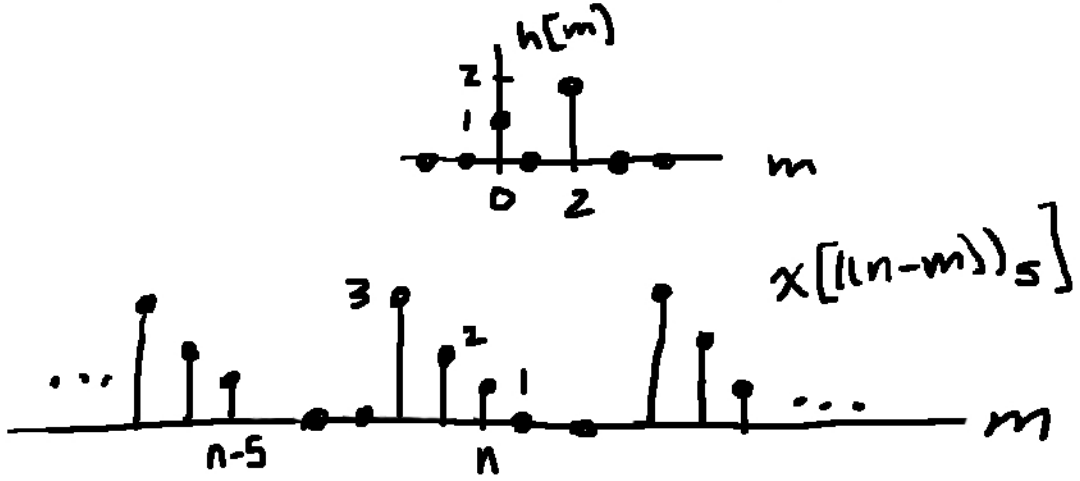
$\tilde{y}[4] = 8$

$\tilde{y}[5] = 5$

$\vdots$

Now consider $h[n] \circledN x[n]$ for $N = 3+3-1 = 5$

$$\tilde{y}[n] = h[n] * x[((n))_5]$$



$\tilde{y}[n] \neq 0 \quad n < 0$

$\tilde{y}[0] = 1$

$\tilde{y}[1] = 2$

$\tilde{y}[2] = 5$

$\tilde{y}[3] = 4$

$\tilde{y}[4] = 6$

$\vdots$

Here $y[n]$ has duration 5 $(y[0], y[1], \cdots y[4] \neq 0)$ and

for $N=5$, $y[n] = \tilde{y}[n]$, $n = 0, 1, 2, 3, 4$

Can recover $y[n]$ from $\tilde{y}[n]$.

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